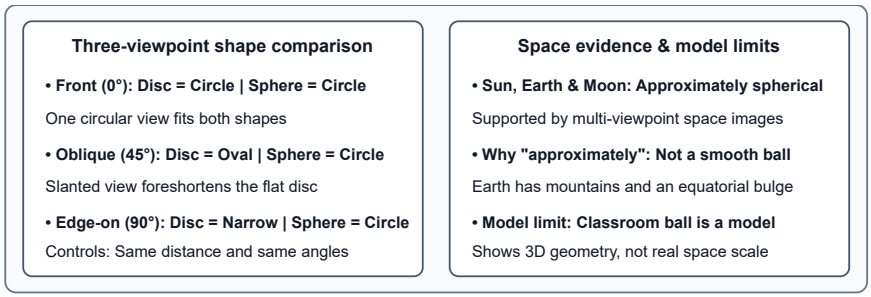
Reference sheets

Reference A · Three-viewpoint comparison: Flat disc vs Solid sphere

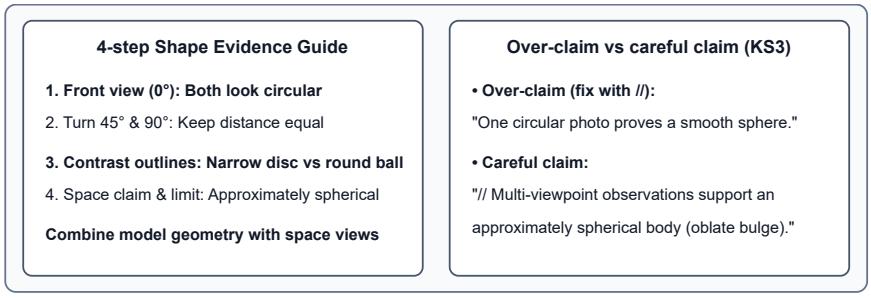


Three-viewpoint shape comparison. It compares a flat disc and a sphere at 0°, 45°, and 90°.

Reference only. Do not write on this sheet. Both objects are held at the same viewing distance across all three viewpoints.

Viewpoint angle.	Flat card disc outline.	Solid sphere (ball) outline.	What this viewpoint tells us.
1. Front view (0°).	Circular outline.	Circular outline.	Both shapes look circular from the front, so one view cannot separate them.
2. Oblique view (45°).	Oval (elliptical) outline.	Circular outline.	A slanted view foreshortens the flat disc while the sphere stays round.
3. Edge-on view (90° quarter-turn).	Narrow thin edge.	Circular outline.	Turning 90° separates the flat disc (narrow) from the 3D sphere (circular).

Reference B · Space evidence, careful claims, and model limits



Four-step Shape Evidence guide. It links 0°, 45°, and 90° views to spacecraft evidence and limits.

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
E1 · Earth from spacecraft.	Spacecraft observations from multiple viewpoints show Earth	Earth has surface mountains and a small equatorial bulge,

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
	as a three-dimensional round globe.	so it is approximately spherical.
E2 · Moon and Sun.	Multi-angle astronomical observations show the Moon and Sun have circular outlines from every viewpoint.	Described as approximately spherical three-dimensional bodies rather than flat discs.
E3 · Classroom ball model.	Predicts how a 3D sphere keeps a circular outline at 0°, 45°, and 90°.	Limit: the classroom ball is a geometric model, not real planetary proof or to scale.

Made by Matt

Tier pack · Supported

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Look at Box 1 on the left of the arrival drawing. Which object is a three-dimensional solid: the ball or the flat circle?

Help. Look at Box 1 on the left.

2. Look at Box 1 again. If a mystery object looks circular from the front (0°), must it be a sphere?

Help. Compare the front view of both objects in Box 1.

3. Look at Box 2 in the middle of the drawing. When the flat disc turns edge-on (90°), does its outline look narrow or wide?

Help. Check the 90° edge-on view in Box 2.

4. Look at Box 3 on the right of the drawing. For a fair comparison, should both objects be held at the same distance?

Help. Check the fair comparison rule in Box 3.

You do tasks

1. Use Reference A. Compare the flat disc and the solid sphere at 0° (front), 45° (oblique), and 90° (edge-on). Write or point to the outline for each view. **Staff I ☐ S ☐ C ☐**

Help. Check Rows 1, 2, and 3 on Reference A.

2. Use Reference A. Complete: 'At 90°, the flat disc looks ___, while the sphere stays ___. We keep viewing distance the ___.' **Staff I ☐ S ☐ C ☐**

Help. Use the words narrow, circular, and same.

3. Use Reference B. Explain why we say Earth, the Moon, and the Sun are 'approximately spherical.' State one limit of the classroom ball. **Staff I ☐ S ☐ C ☐**

Help. Check Steps 3 and 4 on Reference B.

Optional participation-loop prompt. Share your viewpoint comparison and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. When turned edge-on at 90°, which object becomes narrow: a flat disc or a sphere?

2. Why do we say Earth is 'approximately spherical' instead of a perfect smooth ball?

3. Is a classroom ball a helpful model of Earth or a real photograph from space?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Supported

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt